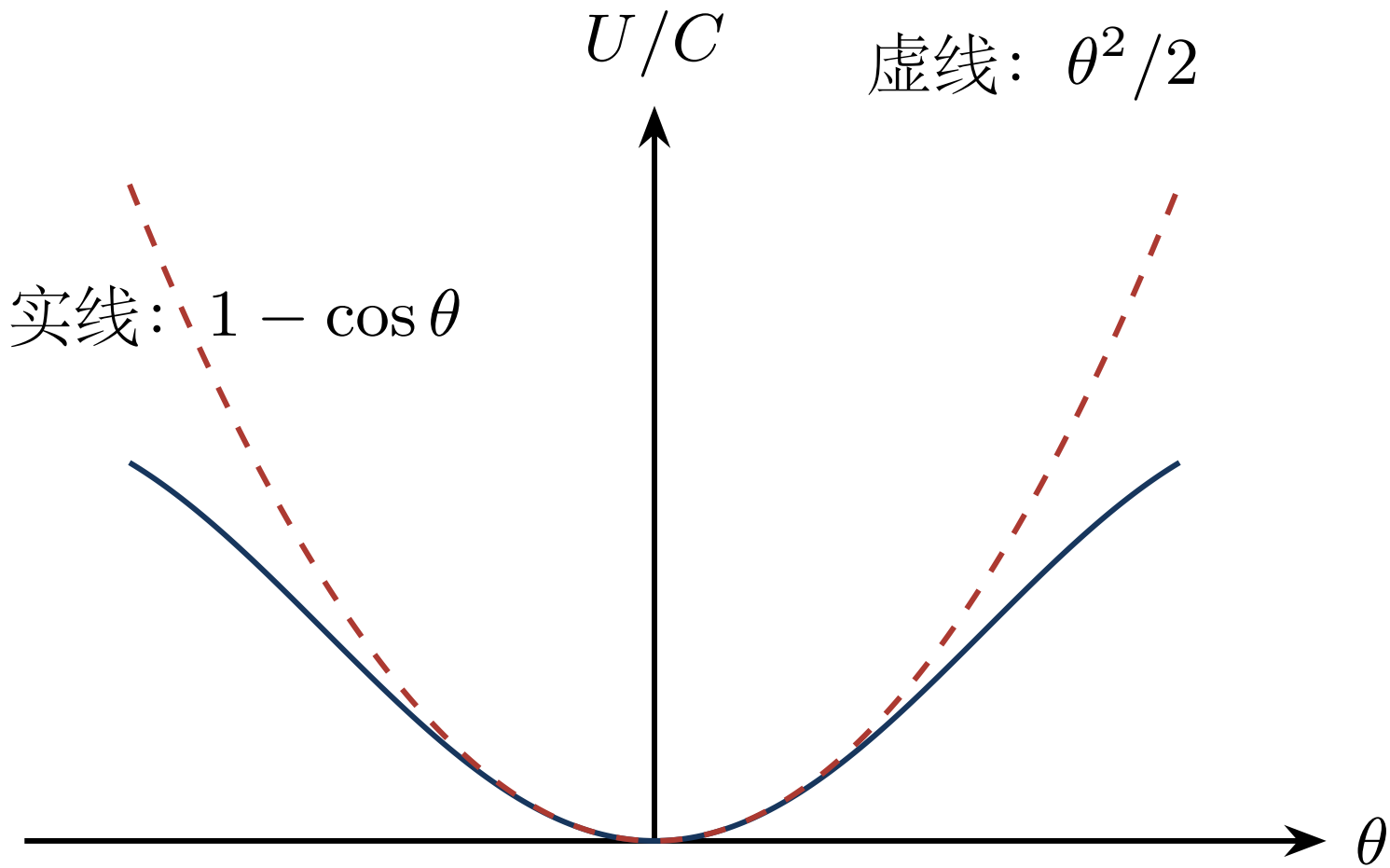

$C = kQld/4$；此势能只针对杆的单一转动自由度。小振动 $\omega^2 = 3kQd/(ml)$。